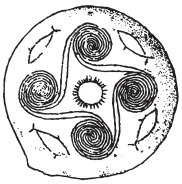
This pdf of your paper in Representations belongs to the publishers Oxbow Books and it is their copyright.

As author you are licenced to make up to 50 offprints from it, but beyond that you may not publish it on the World Wide Web until three years from publication (May 2024), unless the site is a limited access intranet (password protected). If you have queries about this please contact the editorial department at Oxbow Books (editorial@oxbowbooks.com).

AN OFFPRINT FROM

SHEFFIELD STUDIES IN
AEGEAN ARCHAEOLOGY



Representations

Material and Immaterial Modes of Communication in the Bronze Age Aegean

Paperback Edition: ISBN 978-1-78925-641-3
Digital Edition: ISBN 978-1-78925-642-0 (epub)

Edited by
John Bennet

OXBOW | books
Oxford & Philadelphia

Chapter 11

‘Picture-writing’ and phoneticism after *Scripta Minoa* I

Artemis Karnava

The book that constitutes the reference point for this paper is first and foremost a fundamental study on the 2nd millennium BC Aegean writing systems. More than a hundred years ago it clarified a number of basic issues, among these the question of the type to which the Aegean scripts belong: ideographic, syllabic or alphabetic, or a combination of these. In fact, the principle that we are dealing with primarily syllabic writing systems was understood by Evans from the beginning, long before Linear B was deciphered or even first discovered (Evans 1895: 300–2).

Apparently, there is plenty of value that remains from a hundred-year old book, to judge from how many times it is still quoted and with good reason: *Scripta Minoa* I is both a corpus of Cretan Hieroglyphic inscriptions known at the time (*i.e.,* mainly from the Knossos Hieroglyphic Deposit) as well as a synthetic study on this particular writing system, and, in addition, of writing systems in general. To this day, no other book can match it as far as the Cretan Hieroglyphic is concerned. Ferdinand Chapouthier, who excavated the second deposit of Cretan Hieroglyphic documents ever recovered on Crete, in the palace of Malia this time, followed Evans’ lead in his publication (Chapouthier 1930). Yet, if we want to discuss the legacy of *Scripta Minoa* I today in the field of editing of syllabic inscriptions, we find it to be limited: the editing applied to inscriptions recovered in the second half of the 20th century were fundamentally different, in that they offered corpora with the minimum possible degree of interpretation, leaving the reader to work on the inscriptions on his/her own and offer his/her personal interpretative frame (Poursat *et al.* 1978; *CHIC* 1996). Only the most recent publication of a Cretan Hieroglyphic deposit, albeit a relatively small group of documents, seems to retreat to a time when inscriptions were actually discussed in their *editio princeps* (Tsipopoulou and Hallager 2010).

In 1909, therefore, Evans, besides editing the documents in Cretan Hieroglyphic, saw fit to offer an extensive commentary on them. In his treatment of some 160 pages

(Evans 1909: 111–272), he seems never to lose track of what he calls the 'ideographic value' of a '(conventionalized) pictographic or hieroglyphic sign', whereby he concentrates on detecting a pre-existing object prototype behind the form of every sign. He hunts for this presumed 'ideographic' value and his subsequent interpretation is based on how the shape of a sign 'speaks' to his logic and knowledge. In this way, Evans set the stage for the discussion of phoneticism of the Cretan Hieroglyphic script signs, *i.e.,* whether one should see them as phonetic representations, or there was more lurking behind them than mere phonemes.

At times he goes to great lengths in order to justify some of his interpretations, by using some sort of 'free-association' play. An interesting feature is exemplified by his so-called 'cumulative ideographic groups', 'where two or more signs with a closely allied signification are placed together so as to cover a more extended expression' (Evans 1909: 246). Sign-groups on various documents, such as on face **.a** of the seal *CMS* VI 95 (Evans 1909: P. 22a; *CHIC* 1996: #256α) (Fig. 11.1), presumably attest to the notion of the 'Keeper of the Swine', hinted at, in his opinion, by the combination of the 'gate' sign together with the drawing of the animal (Evans 1909: 153).

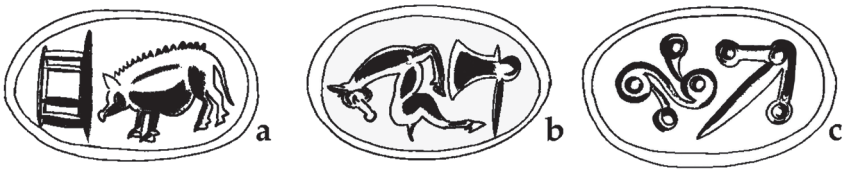


*Fig. 11.1: Three-sided chalcedony prism from Eastern Crete, Ashmolean Museum, Oxford (*CMS *VI 95a;* SM *I 1909: P. 22a;* CHIC *1996: #256α) (drawing by L. Godart in* CHIC *1996).*

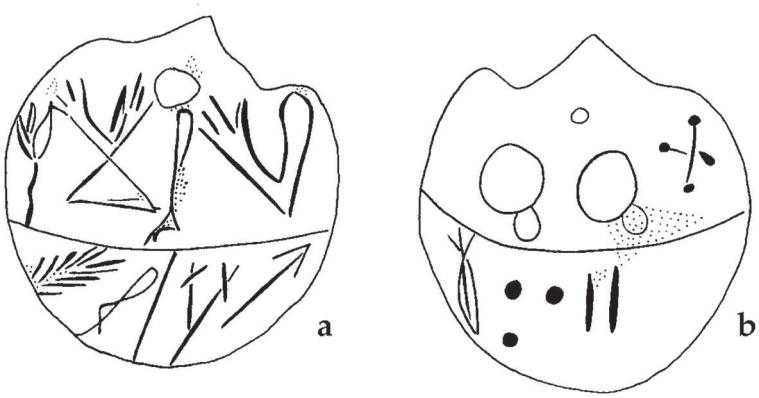


*Fig. 11.2: Medallion with Cretan Hieroglyphic script from the Knossos Hieroglyphic deposit (*SM I *1909: P. 85;* CHIC *1996: #043) (drawing by L. Godart in* CHIC *1996).*

Equally, on a medallion (Evans 1909: P. 85a; *CHIC* 1996: #043a) (Fig. 11.2) the notion of 'labour' is denoted by the combination of a human fore-arm, a leg and crossed arms (Evans 1909: 246).

At the same time, and apparently not realising the interpretative discrepancy, he does make a distinction between signs attested in groups, and signs attested in isolation, signs which we understand today to be of syllabic phonetic value, and signs which stand for a whole word respectively. He also notices a number of instances when the same signs are not only attested within groups, but also on their own, although he finds it hard to explain the phenomenon: 'these hieroglyphs ... were of general use as integral parts of the script' (Evans 1909: 245).

Furthermore, Evans observes that the low overall number of Hieroglyphic signs attested (135 according to his count) points, inescapably, to a phonetic system of writing (Evans 1909: 247). As a corroborative argument he offers the fact that he failed to see any (logical or apparent) connection between signs within the same sign-groups or any 'cumulative value'; he therefore concluded that they could only represent phonetic values. Further on, he compared the numbers of signs found in Aegean sign-groups with the numbers of signs in groups in the Egyptian script and found them to be roughly in proportion. Finally, on more than one occasion, he mentions the notion of 'linear signs', a term which refers purely to the (simple) form of a sign formed after a – high or low – degree of abstraction, which, somehow, was doomed to have a phonetic element (Evans 1909: 1–18, *passim*).

The problem, today

More than a hundred years have elapsed and a number of the issues that Evans was the first to discuss are still unresolved. It is worth noting that the discussion regarding the Cretan Hieroglyphic script nowadays appears to focus intensely on the so-called 'decorative' character of certain Cretan Hieroglyphic script signs on seal faces, a matter that was generated during the making of the inscriptions corpus that we all use today (*CHIC* 1996), as a series of papers by one of the editors exemplifies (Olivier 1981; 1989; 1990; 1995). I was (unfortunately, in retrospect) among the first, to criticise the editorial decision to dismiss certain script signs on seal faces as 'decorative' and thus omit them from the final text transcriptions in the corpus (Karnava 1997). Afterwards, a prolific discussion ensued (*cf.* Jasink 2008; 2009 and multiple papers since; Civitillo 2014; 2016; Decorte 2017), which also occasionally focuses around obscure terms such as 'iconicity' (Ferrara and Christiani 2016; Ferrara and Jasink 2017; Ferrara 2018). Whether this discussion that recycles the same 136 Cretan Hieroglyphic seals and some 57 impressions of Cretan Hieroglyphic seals now lost (the numbers from *CHIC* 1996, but the totals have not changed considerably since), is of interest to anyone outside those who specifically study this particular writing system (attested on some 350 documents in total to date), or can actually produce any meaningful observations, is, to my mind and on the basis of the literature thus far, highly doubtful.

The problem I am interested in however pertains to Evans' belief that the Cretan Hieroglyphic represents the earliest stage in the evolutionary course of writing in 2nd millennium BC Crete, because of its use of 'picture-signs', as opposed to 'linear' signs. Although evolutionism is generally unfashionable nowadays as an archaeological explanation and probably most of us would be unwilling to accept Evans' vague notion of a progression from 'picture-writing' towards an equally vague 'linearism', even if only as a matter of principle, we are still not clear whether the invention of writing in Crete involved some kind of evolution from 'picture-signs' towards syllables, *i.e.*, we remain unclear how Cretan logo-syllabic writing was conceived at its very beginning. This is largely because we know very little of the beginnings of writing in Crete in general. Although a universal progression from 'logography', where 'each symbol represents a complete word or idea', to 'script [with] greater facility in representing the entirety of the language' is still maintained for scripts such as cuneiform, Egyptian and Maya hieroglyphs (Postgate *et al.* 1995: 461), much the same fragmentary picture emerges concerning the beginnings of proto-Elamite, Egyptian, Mesoamerican or even Runic:

> Scattered words; numbers with isolated, probable word-signs or logographs; ancestral forms with unclear tethers to later evidence: the legacy of this period is patchwork, inherently unapproachable because of its paucity of texts, early stages of codification, and sociolinguistic setting that may have evolved exceedingly small numbers of literates or – a more apt word – 'scribal adepts' (Houston 2004b: 349–50).

Writing in the Aegean, even if thought of as an essentially indigenous development, is believed to be the result of the diffusion of an initial 'stimulus'; such a statement is difficult, if not impossible, to prove (Olivier 1986: 378; 1996: 102–4; 2008: 169). Because at or just before the beginning of the 2nd millennium BC, when writing is first attested in the Aegean, Crete had contacts with literate peoples (*i.e.*, Egypt as well as various eastern Mediterranean cultures), one tends to think that it is impossible for writing to have been a totally independent invention. Similarly unsubstantiated claims, 'based on dates and relative physical proximity', have been used in comparable instances (Houston 2004a: 10–11). In the instance of the Aegean scripts, the 'mirage égyptien' – a phrase found in Olivier 1996: 102 – is extremely difficult to shake off, although some cracks have allowed for 'oriental' influences to infiltrate (Powell 2009: 248). In any case, it is highly doubtful to what degree an eventual detection of the provenance of this assumed 'stimulus' would be helpful: the particularities of the adoption/adjustment process would still remain elusive and probably poorly understood. The early stages and development of the Greek alphabet is a similar case in point: although the provenance is agreed upon in scholarship, the technicalities of the process, including the date of adoption, are far from clear (Millard 2012).

It is also a fact that discussion about the same topic, *i.e.*, the beginning of writing, in neighbouring or not-so-neighbouring areas has moved on from the detection of an ultimate 'primary' invention or a 'secondary' adaptation (which appears in any case

inconclusive for now in the specific case of the battle between Egyptian vs. Mesopotamian supremacy, Baines 2004: 175–7), to discussions that centre around notions such as the complementary purposes of 'communication' as well as 'display' (Baines 1989), and the antithetical 'utilitarian' or 'ceremonial' functions (Postgate *et al.* 1995). That said, there still appears to exist space for cautiously suggested evolutionary explanations, which function in the context of an extremely *longue durée*, such as in the instance of China (Neolithic signs that anticipate later Chinese characters: Li *et al.* 2003), of Egypt (the early script formation period spun some 500 years: Regulski 2010), of Mesopotamia (clay 'tokens' and 'envelopes' as precursors to writing: Woods 2010).

In the Aegean Bronze Age environment, the evidence remains too scarce to allow for such elaborate interpretations. Since we seem to be struggling with elementary matters, I focus on one of these in this paper. The discussion, initiated by Evans over a century ago, will focus on what is described here as the symbiotic relationship between syllabograms and logograms, both categories of signs that apparently existed already in what we know to be the earliest attested stage of writing in Crete. Progress on this issue has certainly been achieved in relation to the later Linear B script since the decipherment and as a result of it, but our knowledge of the beginning(s) of this process remains rather poor.

The beginning

Along with most scholars, I believe, I take as the very first evidence for writing in Crete what is commonly – and misleadingly – known as the 'Archanes script'. The term was coined by Paul Yule is his study on early Cretan seals. Yule referred to a group of seals distinguished, in his opinion, 'primarily by the signs represented'; he treated these 'signs' as 'motifs' (the 'dentate band', 'leaves', 'double axe', 'ewers' and 'cuts shaped like Vs'), which usually appeared in a field bordered by an incision. Some 15 specimens were listed under this name; the 'motifs' appeared on a limited variety of seal shapes (flat-sided discs, discoids with bordered motifs, gables and reels). Additionally, the materials used for their manufacture were either 'serpentine' or 'ivory'. The production of these seals not only falls into a specific time frame (EM III-MMIA), but, in Yule's treatment, they fell into a specific style-group, namely, the 'Border/Leaf Complex' (Yule 1980: 170). The 'Border/Leaf Complex' comprised some 160 specimens, found from Archanes and Gournes in the Pediada region of north-central Crete, to Lendas in the Mesara in the south (Yule 1980: 209–10). Yule's estimate that the 'earliest clearly definable script known in Crete may be called the Archanes script' is misleading, however, since it assigns to the 'Archanes script' the status of a writing system apart from the already known Protopalatial scripts (Cretan Hieroglyphic and Linear A), which simply is not the case.

In scholarship produced by researchers of early writing in Crete, the term which is most commonly used is the 'Archanes formula' (*CHIC* 1996: 18, n. 59–60; Poursat 2000). The seals listed by Yule as belonging to the 'Archanes script' constitute a larger

group than those examples that contain the 'Archanes formula'. Evans used the term 'formula' interchangeably with the term 'sign-group', without any specification of either meaning (Evans 1909: 260–72; Grumach 1969: 235). His belief that some of the signs in sign-groups had a non-phonetic value led him to conclude that sign-groups did not stand for words containing signs of syllabic value, but that they conveyed a symbolic meaning, which he interpreted as a mixture of personal names and titles. Because, however, sign-groups consisted of repetitive combinations of signs (which is one of the reasons why it is proven that they are signs in a writing system), he referred to sign-groups as 'formulas'.

Here I introduce the term 'the Archanes inscription', in order to assert that I interpret it as an attestation of writing and thus of wording. It is actually the standard combination of two sign-groups consisting of four different signs, attested as a whole or in part on 14 different occasions.[1] It remains uncertain, however, to which Protopalatial writing system this set of sign-groups should be attributed: Cretan Hieroglyphic or Linear A, although there appears little doubt that it belongs to one of the two (or both). The majority of researchers seem to take it for granted that the inscription was written in the Cretan Hieroglyphic. The arguments for its plausible listing among (?) Linear A documents are put forward by Godart (1999; *pace*, Karnava 2000: 195–8, 216–17), and the problem is succinctly laid out in Olivier (2008: 170–2). The most recent discussion adds a further a seal to the meagre 'Archanes inscription' corpus (Perna 2019, with previous bibliography).

The name bears an association with the site of Archanes, because the cemetery of Phourni at Archanes constitutes the primary find-spot of seals inscribed with the inscription; other find-spots of seals bearing the 'Archanes inscription' include Knossos and Moni Odigitria on Crete (Sbonias 2010: S35). It is also attested on impressions (of seals which have never been discovered) at Knossos and on the island of Samothrace, at the northern end of the Aegean Sea. The seals, at least the ones retrieved in Archanes, have been dated by their excavator as early as the MM IA period (*i.e.,* the latest Pre-Palatial phase), whereas the impressions presumably fall within MM II, or broadly speaking, the Proto-Palatial period (Poursat, in *CHIC* 1996: 27–32).

What I would like to isolate from this small sample of seals is the well-known bone (previously published as ivory, corrected by Krzyszkowska 1989: 123; Sbonias 1995: 59, Class 3) seal described as a '4-sided bar', an *unicum* as far as the shape is concerned (Yule 1980: 100, subclass 34a), as it is in many respects. It has 14 faces (three on each long side of the 'bar' and two on the short sides), two of which bear the 'Archanes inscription' (*CMS* II1 391; *CHIC* 1996: #315). It was found in the Phourni cemetery, in burial Building 6, and its context is dated to MM IA. I have used this seal evidence elsewhere (Karnava 2005) to suggest that a third face of this seal (face G), on the same side as the 'Archanes inscription' (faces I and H), can be paralleled in an inscription on a pottery sherd retrieved from the Levantine site of Tel Haror. I also suggested that a possible presence of logograms on face G would make the two inscriptions

Fig. 11.3: Faces G, H, I, L and J of the bone seal CMS *II 1 391, as drawn by L. Godart (courtesy of L. Godart; only faces I and H appear in* CHIC *1996, faces G, J and L to be seen in Karnava 2005, pl. CXCVIIIb). The drawing of the remaining faces (A-F, K, M and N) come from* CMS *(after* CMS *II 1 391 and Karnava 2005: pl. CXCVIIIb) (*CMS *images © Institute for Classical Archaeology, University of Heidelberg).*

(the 'Archanes inscription', as attested on the Archanes bone seal, and the Tel Haror inscription) appear more compatible (Fig. 11.3).

In the present context the special interest of this object is not so much the inscription itself, as the accompanying motifs on the remaining seal faces. The obvious question is: what other inscriptions existed at the beginning of Cretan writing besides the 'Archanes inscription'? Some of the other faces on this particular object may give us an indication. It can be suggested with a degree of confidence, that this object, although of an early date (*i.e.,* earlier than the majority of Cretan Hieroglyphic

seals) fits very well within the Hieroglyphic Proto-Palatial environment, and with the canonical and undoubtedly Cretan Hieroglyphic seals, for a number of reasons. Apart from five motifs depicting whole animals (faces A-C, E, N), an unidentified object which seems to be a manufactured item (face F), a human figure in the process of doing/carrying something (face K), and a schematic flower (face M), some of the remaining motifs are reminiscent of Hieroglyphic signs known from later documentation, namely a 'sistrum' (face D, corresponding to Hieroglyphic sign **181*), a human 'hand' (face J, sign 008) and a human 'leg' (face L, sign 010).

The two motifs on faces J and L (the 'hand' and the 'leg' respectively) are non-existent as what one would call decorative seal motifs but quite at home within the Cretan Hieroglyphic sign repertoire, especially in sign-groups attested on seals. The rest of the evidence shows that when a 'hand' or a 'leg' is attested on a seal, it is part of a Cretan Hieroglyphic sign-group: they never appear as parts of random decorative motifs and they disappear along with the Cretan Hieroglyphic seals. Only recently a specimen of an isolated 'hand' representation came to light on an impression on an administrative document from Knossos dated to MM IIA (*CMS* II 8 15; Weingarten 2003; 2007). Although this single 'hand' does not by any means constitute an inscription in the strict sense of the word (*i.e.*, two or more signs together), it is tempting to see it as a motif derived from a literate environment, which undoubtedly existed in Knossos in MM IIA.

The motif on face D (the 'sistrum') corresponds perfectly with the Cretan Hieroglyphic sign **181*, which is attested once on a Cretan Hieroglyphic seal and is transcribed as a logogram (*CMS* IV 136d; *CHIC* 1996: #305δ) (Fig. 11.4). On the same face as sign **181* we encounter another sign interpreted as a logogram (sign **180*); this second sign however is also attested on two Cretan Hieroglyphic 4-sided bars from Knossos (#065.c and #067.b). The same seal bears on two other faces (.β and .γ respectively) two of the most frequently attested Cretan Hieroglyphic sign-groups, 044–049 and 044–005 (the 'trowel-arrow' and the 'trowel-eye' sign-groups, as Evans would have described them: *e.g.* Evans 1909: 260).

Is all this evidence circumstantial? Logograms on Hieroglyphic seal faces, for reasons we cannot quite grasp, are not altogether unknown (see, for instance, *CMS* III 149; *CHIC* 1996: #206) (Fig. 11.5). One would perhaps expect logograms, signs intended to serve the administration and so strictly connected to the counting and controlling of the movement of products, to have no purpose on seals. Yet, in my opinion, their presence constitutes a confirmation of how closely the manufacture and use of Cretan Hieroglyphic seals was to Proto-Palatial administrative procedures.

What conclusions can be drawn from the previous observations? Firstly, that signs which transmitted the significant segments of speech, namely words, appear to have existed in the writing system all along. From the very beginning (or at least from the earliest attested examples), signs stand as the single motif on a seal face, and they are signs that are drawn from a Cretan Hieroglyphic sign repertoire already in existence at the time. Whether they stood for either a 'purely visual symbol' or 'a syllabic

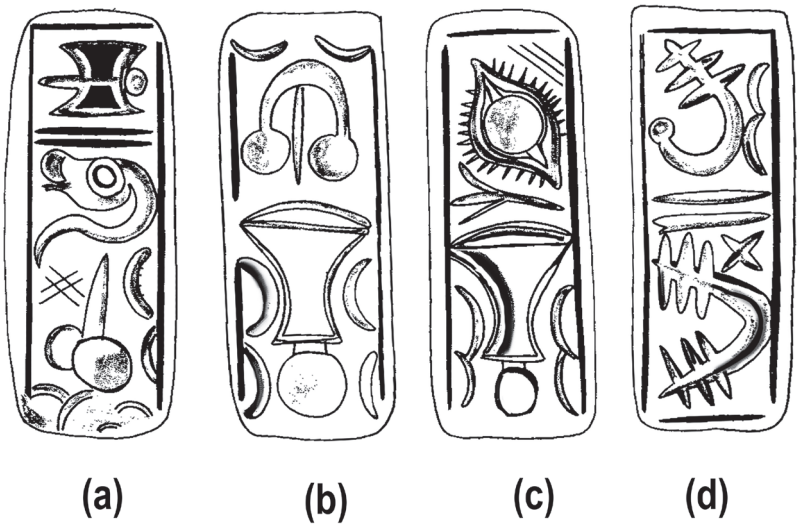


*Fig. 11.4: Four-sided jasper prism from Lastros, Crete, in the Metaxas collection/Herakleion Museum (*CMS *IV 136d;* CHIC *1996: #305d) (drawing by L. Godart in* CHIC *1996).*

sign used in abbreviation' according to Ventris' and Chadwick's definition of logograms (Ventris and Chadwick 1973: 48), they are logograms, or at least they fit our accepted definition of logograms, as suggested for the Linear B evidence and, by extension, for the other two scripts.

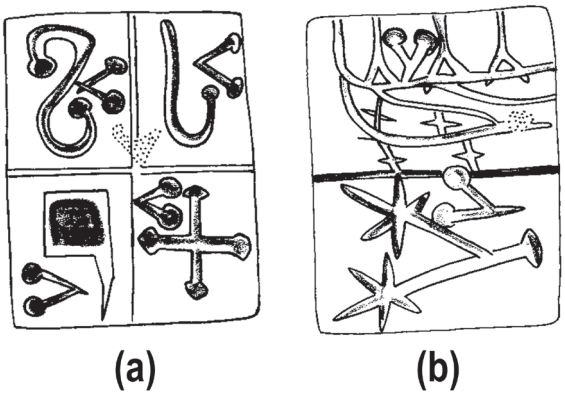


*Fig. 11.5: An onyx flattened cylinder from Crete in the Giamalakis collection/Herakleion Museum (*CMS *III 149;* CHIC *1996: #206) (drawing by L. Godart in* CHIC *1996).*

The objection that these signs could be part of the decorative repertory which ultimately inspired the creation of the Cretan Hieroglyphic script afterwards does not stand: these signs are attested in conjunction with Cretan Hieroglyphic seals from their earliest examples and they disappear as motifs together with the end of the manufacture of Hieroglyphic seals. This suggests the opposite trajectory, namely that the seal engraving repertory borrowed from the Cretan Hieroglyphic repertory, but as the script died out, so did the borrowing.

Another objection, that logograms, when they appear on the proper archival documents are supposedly meant to be counted (Ventris and Chadwick 1973: 48), and they

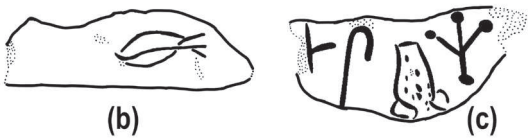


*Fig. 11.6: A Cretan Hieroglyphic crescent from Knossos, with notation of the 'grain' logogram (side b), but no numerical entries (*SM *I 1909: P. 52;* CHIC *1996: #017) (drawing by L. Godart in* CHIC *1996).*

are thus completely out of place on a seal face, is also not valid in the Cretan Hieroglyphic. A series of true Cretan Hieroglyphic archival documents, the sealings called 'crescents', attests to the use of logograms without numerals, *i.e.*, logograms that were not meant to count anything, logograms that are just there as an indication of a content of some sort (Fig. 11.6). Even logograms on roundels in the Linear A writing system were meant to be counted, just by means of repetitive seal impressions (*cf.* Hallager 1996).

A second thought coming from the previous observations is that from this very early stage in the history of writing in Crete certain signs seem to hint at their double role, the one they will assume with certainty later on in all three scripts: as both syllabograms and logograms. Even if we are willing to dispute the patchy evidence offered by the unique Archanes bar, seals of the Proto-Palatial period come to our aid: the 'fig', which we see as a logogram, is also attested as a syllabogram in the Cretan Hieroglyphic script (see Fig. 11.5b above).

Linear B

It appears that the dual function of signs as syllabograms *and* logograms, a feature attested from the very beginning of the history of writing in the 2nd millennium BC Aegean, persists throughout. The Linear B script, which represents the opposite end of the chronological spectrum to the 'Archanes inscription', is interesting in that it demonstrates that the same process is still at work. The original classification of logograms in Linear B was offered according to the way they were created (Ventris and Chadwick 1973: 48–53). They were classified into six categories and the reader was warned that the different origin of logograms did not affect their status:

1. naturalistic and self-explanatory pictorial signs (such as the logograms for 'horse', 'chariot', 'vase');
2. conventionalised and abstract pictorial symbols (recognised mostly when spelled out phonetically, such as *pa-we-a, ka-ko, e-ra$_3$-wo*);
3. ideograms ligatured with a syllabic sign (animal hide+KO [logogram *153], animal hide+WE [logogram *152] etc.);
4. single syllabic signs used as ideograms (deriving from the initials of Greek [or non-Greek?] words);
5. 'monograms' (two or three syllabic signs 'telescoped');
6. 'adjuncts' (small syllabic signs written before ideograms).

When these categories are examined closely, one realises that we are in fact dealing with two major underlying explanations as to how Linear B logograms came into being: the first explanation is that one should take the image of the object as such ('pictorial' signs) and the other is that we are dealing with phonetic signs used acrophonically. Caution is necessary, however, in that the so-called 'pictorial' ideograms were probably not as realistic as their details would have us believe, because verbally expressed information in the accompanying texts contradicts at times the 'pictorial' information (Ventris and Chadwick 1973: 391; Weilhartner 2015; Petrakis 2017).

The latter major group of logograms, that of 'syllabic' logograms, has a name, which inherently implies that the phonetic transcription of the word had taken place before the creation of the corresponding abbreviated logogram (Palaima 1987: 506): a concept such as acrophony demands an advanced degree of realisation of the language structures and how these can be used for the sake of written communication. Acrophony is also suggested, but equally difficult to prove, in the instance of the creation of the Greek alphabet (Willi 2008: 407–10). At least in the case of Linear B, this observation undermines any theories of the development of the script from picture-signs towards phoneticism: what we are seeing is exactly the opposite: *i.e.*, a script with a strong phonetic element and logograms created secondarily according to scribal needs, but based on the *phonetic* value of a given sign.

It is essential to bear in mind the numbers of signs that form the basis of our discussion. While all three Aegean scripts have a repertory of more or less 80–100 phonetic signs, in Linear B the most recent estimate gives us 121 logograms, Linear A attests to 84 logograms, and this total includes only the simple forms, not ligatures or those that indicate the sex of an animal (Lebessi *et al.* 1995: 65, n. 8). The number of logograms attested in the Cretan Hieroglyphic script texts is small, when compared to the respective numbers in Linear A and Linear B: there are 33 different signs defined as logograms and 19 of these (to give an indication of the scarcity of the material) are attested only once (*CHIC* 1996: 13, 17, 423–9). Although our evidence is compromised by the scarcity of data, the numbers alone suggest a progression towards a larger population of logograms in the latest of the three scripts. But a rough estimate of 'syllabic' logograms gives a count of 10 among the plentiful Linear B evidence, and another 10, among only 33, in Cretan Hieroglyphic, that is a proportionally higher figure. For reasons to be explained below, I pass over the Linear A evidence for now, because it is infinitely more complicated.

The mechanism that we detect at play in the Proto-Palatial Cretan Hieroglyphic script is apparently still at work some 500 years later, and it is this mechanism in Linear B that predominantly creates images and fewer syllabic symbols. In this framework, the Cretan Hieroglyphic appears more at ease with switching between image and text than is Linear B.

A number of crucial pieces of information regarding the formation and explanation of the various logograms require setting out here (Ventris and Chadwick 1973: 48–53).

The overall explanation for ligatured ideograms, for instance, is that the syllabic sign which is 'surcharged' above or inside the ideogram is the initial of a Greek noun or adjective; the confirmation for this suggestion comes when the noun or adjective is also spelled out phonetically. Nonetheless, a problem already observed is that this practice appears to have been inherited from Linear A, which most scholars agree notated a different language, and some ligatures were inherited directly (*e.g.* TELA+ZO, TELA+KU). So, in Linear B we assume that only part of the mechanism to create logograms was inherited: another part concerned a script-based imagery which was so potent as to induce Linear B creators and modifiers to adopt blindly not only logograms (such as 'wine' and fig'), but also ligatured logograms with, in all probability, nonsensical meanings in the environment of the Greek language.

Linear A

Linear A in turn appears to represent a more complex state of affairs than the other two writing systems. In the Linear A corpus, signs of presumed phonetic value are not distinguished by logograms (*GORILA* V: xviii-xix). In a coarse classification of signs we find under rubric 'A' the so-called 'isolated signs'. In Linear B terms, we would immediately think of logograms; however, because these signs are further divided into six categories, there are at least four which interest us: (1) completely isolated; (2) 'monosyllabic words'; (3) 'transaction signs'; (4) 'logograms'. Moreover, the readers are further warned that 'the sign, even if surrounded by other signs on a certain document, does not necessarily form part of a 'sign-group' in the strict sense of the word' (*GORILA* V: 138).

When going through the attestations of Linear A signs, we realise that overwhelming majority of their use was as part of sign-groups. Therefore, we assume, with a degree of caution, that they had a phonetic value. However, at one point or other, in addition they exhibit a logographic behaviour, either each sign on its own, or as parts of the numerous ligatures, the apparent predecessors of the Linear B 'telescoped' logograms (see above). Very few signs escape the rule and seem to have had a clear-cut role either solely as phonetic signs or solely as logograms. This observation points once more to a writing system that is preponderantly phonetic, because it allows for such an inter-play between sign functions.

Comparanda

To return to Evans, briefly, and in his defence: it is admittedly difficult to lose track of the fact that the signs of the Aegean syllabic scripts – of the Cretan Hieroglyphic and of Linear A mostly – are vividly reminiscent of animate or inanimate objects. There are, however, a couple of instances, where the same train of thought was used to produce, on the one hand, a relatively modern treatment of the Egyptian Hieroglyphic script, and, on the other, an artistic outcome.

The first instance concerns the book *Hieroglyphics* by Horapollo, a person who it is thought was active in the 5th century AD. His book was 're-discovered' by the traveller Cristoforo Buondelmonti around 1420 on the island of Andros, in the form of a manuscript: he found two books 'written in the Egyptian language and translated into Greek by a certain Philip' (Pope 1999: 11, 18–21). This book was subsequently translated into Latin and with the invention of typography, it became tremendously popular (Pope 1999: 27). It also influenced – or, rather, hampered – the decipherment of the Egyptian Hieroglyphic script (Faivre-Martin 2009: 188–9). Thinking that an ancient author probably had good knowledge of the writing system, would-be decipherers depended heavily on the information it contained for clues. Yet, Horapollo's treatment, most likely written when all knowledge of the Egyptian hieroglyphic had already vanished, relies primarily on naïve explanations of the signs as pictures, and was ultimately of little help to the decipherment. Some of these clues run as follows:

> I 23 To indicate a man who has never travelled they paint a man with a donkey's head. For he never knows or listens to accounts of what happened abroad (Pope 1999: 18–19).

The second instance comprises the so-called 'talismanic' seals, manufactured in the Neopalatial period, probably slightly after the Cretan Hieroglyphic went out of use. These seals, which, from the point of view of their craftsmanship are the descendants of hard stone Proto-Palatial seals (and they, in their turn, include some of the finest examples of Cretan Hieroglyphic seals) present us with repetitive decorative motifs; some of these were also part of the Cretan Hieroglyphic signary in the previous period (Onassoglou 1985: 187–9). The fact, however, that these motifs are monotonously and randomly repeated, leads us to think that all knowledge of the Cretan Hieroglyphic script had vanished in the Neopalatial period. Craftsmen do not draw signs to carve from a fully functional signary, such as the one we see behind Hieroglyphic seals: now they seem to repeat earlier motifs, probably 'traditional' at their time, with no knowledge that these 'motifs' were once bearers of phonetic values (Karnava 2000: 207–8).

Much like Horapollo and the craftsmen manufacturing 'talismanic' seals, Evans saw pictures, and he never managed to leave aside unfounded interpretations of his sign-groups, although his evidence pointed elsewhere and he was the first to note it.

Conclusions

In our times, a more thorough analysis of the structural elements of the logo-syllabic Aegean scripts constructs a different picture than that described by Evans. The earliest available evidence of writing in Crete demonstrates the use of both syllabograms and logograms from the very start. Moreover, it shows the use of logograms which could be defined, future finds permitting, as 'syllabic', pointing to a preponderantly phonetic perception of the essence of ancient writing. This point has also been made

independently by Powell (1999) and forms the basis of his theoretical approach to the history of writing.

The fact that all three logosyllabic Aegean scripts, from their appearance to their disappearance, present the phenomenon of interaction between images and text, shows on one hand their common lineage and, on the other, a minimal degree of evolution. With the shreds of evidence we have in our hands, it seems that the Linear B scribes were more interested in introducing images as representations of objects than had those who wrote in the earliest scripts, a fact that appears to demonstrate an opposite course of development than the one suggested to date: we seem to be moving from the preponderance of phoneticism of the Cretan Hieroglyphic script to the invention of new 'picture-signs' for a number of new administrative needs served by the Linear B script. This may also make us suspicious as to whether these scripts were ever used for anything other than product lists: if Linear B novelties and interventions in the writing system consist of new images, which have a limited potential for transmitting abstract ideas and meaning, it is highly doubtful that the script ever recorded other varieties of speech and/or text. This is not to say that poems, for instance, or other, somewhat more complicated verbal expressions than catalogues, did not exist; they were probably limited to the realm of orality (Bennet 2001: 34) and were neither destined to be nor needed to be recorded by means of writing.

Acknowledgements

I would like to thank the organiser of the Round Table, John Bennet, for the invitation to participate in the Sheffield Round Table in 2009, when the core version of this paper was delivered and submitted, additionally because his and his family's extraordinary hospitality remain memorable to this day. Thanks are also due to Carl Knappett and the late Vincenzo La Rosa for putting at my disposal at the time their work prior to publication.

Note

1. The Hieroglyphic signs that appear in the 'Archanes inscription' (numbers taken from *CHIC* 1996: 17): 042-019/019-095-052. The seals are the following (their enumeration following again *CHIC* 1996): #134, #135, #136, #137, #179, #201 (?),#202, #203, #205, #251, #252, #292, #313, #315. For a short explainer, see CHIC 1996: 18, n. 59–60.

Abbreviations

CHIC Corpus Hieroglyphicarum Inscriptionum Cretae (Études crétoises 31). J.-P.Olivier and L. Godart, avec la collaboration de J.-Cl. Poursat. 1996. Paris: De Boccard.

CMS Corpus der minoischen und mykenischen Siegel. F. Matz *et al.* 1964–2018. Berlin: Mann.

GORILA Recueil des inscriptions en linéaire A I-V (Études crétoises). L. Godart and J.-P. Olivier. 1976–1985. Paris: Librairie Orientaliste Paul Geuthner.

SM I *Scripta Minoa* I. A.J. Evans. 1909. Oxford: Clarendon Press.

Bibliography

Baines, J.
1989 Communication and display: the integration of early Egyptian art and writing. *Antiquity* 83: 471–82.
2004 The earliest Egyptian writing: development, context, purpose. In S.D. Houston (ed.), *The First Writing. Script Invention as History and Process*: 150–89. Cambridge: Cambridge University Press.

Bennet, J.
1999 Agency and bureaucracy: thoughts on the nature and extent of administration in Bronze Age Pylos. In S. Voutsaki and J. Killen (eds.), *Economy and Politics in the Mycenaean Palace States. Proceedings of a Conference held on 1–3 July 1999 in the Faculty of Classics, Cambridge*: 25–37. Cambridge: Cambridge Philological Society.

Chapouthier, F.
1930 *Les écritures minoennes au palais de Mallia* (Études crétoises 2). Paris: Librairie Orientaliste Paul Geuthner.

Civitillo, M.
2014 Le strategie communicative sui sigilli con iscrizioni in geroglifico minoico. *Pasiphae* 8: 23–44.
2016 *La scrittura geroglifica minoica sui sigilli: il messaggio della glittica protopalaziale* (Biblioteca di Pasiphae, Collana di filologia e antichità egee 12). Pisa and Roma: Fabrizio Serra.

Decorte, P.-J.E.R.
2017 Cretan 'Hieroglyphic' and the nature of script. In P.M. Steele (ed.), *Understanding Relations Between Scripts: the Aegean Writing Systems*: 33–56. Oxford and Philadelphia: Oxbow.

Evans, A.J.
1895 Primitive pictographs and a prae-Phoenician script, from Crete and the Peloponnese. *Journal of Hellenic Studies* 14: 270–372.

Faivre-Martin, É.
2009 Jean-François Champollion et les hiéroglyphes égyptiens. In B. Lion and C. Michel (eds.), *Histoires des déchiffrements. Les écritures du Proche-Orient à l'Égée*: 187–97. Paris: Errance.

Ferrara, S.
2018 From icon to sign. *Terrain* 70 (url: http://journals.openedition.org/terrain/17225).

Ferrara, S. and D. Cristiani
2016 Il geroglifico cretese: nuovi metodi di lettura (con una nuova proposta di interpretazione del segno 044). *Kadmos* 55: 17–36.

Ferrara, S. and A.M. Jasink
2017 To have and to hold: Hieroglyphic seals as personal markers and objects of display. In A.M. Jasink, J. Weingarten and S. Ferrara (eds.), *Non-scribal Communication Media in the Bronze Age Aegean and Surrounding Areas: the semantics of a-literate and proto-literate media (seals, potmarks, mason's marks, seal-impressed pottery, ideograms and logograms, and related systems)* (Periploi – Collana di Studi egei e ciprioti 9): 41–53. Firenze: Firenze University Press.

Godart, L.
1999 L'écriture d'Arkhanès: hiéroglyphique ou linéaire A? In P.P. Betancourt, V. Karageorghis, R. Laffineur and W.-D. Niemeier (eds.), *MELETEMATA. Studies in Aegean Archaeology presented to Malcolm H. Wiener as he enters his 65th year.* (Aegaeum 20): 299–302. Liège and Austin: University of Liège and University of Texas at Austin.

Grumach, E.
1969 Die kretischen und kyprischen Schriftsysteme. In U. Hausmann (ed.), *Handbuch der Archäologie*: 234–88. München: Beck.

Hallager, E.
1996 *The Minoan Roundel and Other Sealed Documents in the Neopalatial Linear A Administration,* Volumes I-II (Aegaeum 14). Liège and Austin: Peeters.

Houston, S.D.
2004a Overture to *The First Writing*. In S.D. Houston (ed.), *The First Writing. Script Invention as History and Process*: 3–15. Cambridge: Cambridge University Press.
2004b Final thoughts on first writing. In S.D. Houston (ed.), *The First Writing. Script Invention as History and Process*: 349–53. Cambridge: Cambridge University Press.

Jasink, A.M.
2008 The 'scroll' and its variants: from a decorative symbol to a writing sign. In A. Sacconi, M. Del Freo, L. Godart and M. Negri (eds.), *Colloquium Romanum: atti del XII colloquio internazionale di micenologia, Roma, 20 - 25 febbraio 2006*, 1 (Pasiphae 1): 377–93. Pisa and Roma: Fabrizio Serra.

Jasink, A.M. (with contributions by J. Weingarten and L. Bombardieri)
2009 *Cretan Hieroglyphic Seals: a New Classification of Symbols and Ornamental/filling Motifs* (Biblioteca di Pasiphae. Collana di filologia e antichità egee 8). Pisa and Roma: Fabrizio Serra.

Karnava, A.
1997 Review of *CHIC* 1996. *Studi Micenei ed Egeo-Anatolici* 38: 300–4.
2000 *The Cretan Hieroglyphic Script of the Second Millennium BC: description, analysis, function and decipherment perspectives.* PhD Thesis, Université Libre de Bruxelles.
2005 The Tel Haror inscription and Crete: a further link. In R. Laffineur and E. Greco (eds.), *EMPORIA. Aegeans in Central and Eastern Mediterranean, Proceedings of the 10th International Aegean Conference/, Athens, Italian School of Archaeology, 14–18 April 2004* (Aegaeum 25): 837–43. Liège: University of Liège.

Krzyszkowska, O.
1989 Early Cretan seals. New evidence for the use of bone, ivory and boar's tusk (*CMS* Beiheft 3): 111–26. Berlin: Gebr. Mann.

Lebessi, A., P. Muhly and J.-P. Olivier
1995 An inscription in the Hieroglyphic script from the Syme sanctuary, Crete (SY Hf 01). *Kadmos* 34: 63–77.

Li, X., G. Harbottle, J. Zhang and C. Wang
2003 The earliest writing? Sign use in the seventh millennium BC at Jiahu, Henan province, China. *Antiquity* 77: 31–44.

Millard, A.
2012 The alphabet. In H. Gzella (ed.), *Languages from the World of the Bible*: 14–27. Boston/Berlin: de Gruyter.

Olivier, J.-P.
1981 Les sceaux avec des signes hiéroglyphiques. Que lire? Une question de définition (*CMS* Beiheft 1): 105–15. Berlin: Gebr. Mann.
1986 Cretan writing in the second millennium BC. *World Archaeology* 17: 377–89.
1989 The possible methods in deciphering the Pictographic Cretan script. In Y. Duhoux, T.G. Palaima and J. Bennet (eds.), *Problems in Decipherment* (*BCILL* 49): 39–58. Louvain-la-Neuve: Peeters.
1990 The relationship between inscriptions on Hieroglyphic seals and those written on archival documents. In T.G. Palaima (ed.), *Aegean Seals, Sealings and Administration. Proceedings of the NEH-Dickson Conference of the Program in Aegean scripts and Prehistory of the Department of Classics, University of Texas at Austin, January 11–13, 1989* (Aegaeum 5): 11–24. Liège: University of Liège.

1995 Les sceaux avec des signes hiéroglyphiques. Que lire? Une question de bon sens (*CMS* Beiheft 5): 169–81. Berlin: Gebr. Mann.

1996 Les écritures crétoises: sept points à considerer. In E. De Miro, L. Godart and A. Sacconi (eds.), *Atti e memorie del IIo Congresso Internazionale di Micenologia, Roma-Napoli, 14–20 Ottobre 1991*, I: 101–13. Rome: Gruppo Editoriale Internazionale.

2008 Chapitre II. Les écritures crétoises. In R. Treuil, P. Darcque, J.-Cl. Poursat and G. Touchais, *Les civilisations égéennes, du Néolithique et de l'Âge du Bronze*: 169–83. Paris: Presses Universitaires de France.

Palaima, T.G.

1987 Comments on Mycenaean literacy. In J.T. Killen, J.L. Melena and J.-P. Olivier (eds.), *Studies in Mycenaean and Classical Greek presented to John Chadwick* (Minos 20–22): 499–510. Salamanca: University of Salamanca.

Perna, M.

2019 A seal in the British Museum with a Cretan Hieroglyphic inscription (CR (?) S (1/1) 07). *Kadmos* 58: 49–59.

Petrakis, V.

2017 Figures of speech? Observations on the non-phonographic component in the Linear B writing system. In M.-L. Nosch and H. Landenius Enegren (eds.), *Aegean Scripts. Proceedings of the 14th International Colloquium on Mycenaean Studies, Copenhagen, 2–5 September 2015* (Incunabula Graeca 105.1): 127–67. Rome: Consiglio Nazionale delle Ricerche Edizioni.

Pope, M.

1999 *The Story of Decipherment from Egyptian Hieroglyphs to Maya Script*. Revised edition. London: Thames and Hudson.

Postgate, N., T. Wang and T. Wilkinson

1995 The evidence for early writing: utilitarian or ceremonial? *Antiquity* 69: 459–80.

Poursat, J.-Cl.

2000 Les sceaux hiéroglyphiques dans l'administration minoenne: usage et fonction. In M. Perna (ed.), *Administrative Documents in the Aegean and their Near Eastern Counterparts: Proceedings of the International Colloquium, Naples, February 29-March 2, 1996*: 187–91. Torino: Centro internazionale di ricerche archeologiche, antropologiche e storiche.

Poursat, J.-Cl., L. Godart and J.-P. Olivier

1978 *Fouilles exécutées à Mallia. Le Quartier Mu* (Études crétoises 23). Paris: Librairie Orientaliste Paul Geuthner.

Powell, B.B.

2009 *Writing. Theory and History of the Technology of Civilization*. Malden-Oxford: Wiley-Blackwell.

Regulski, I.

2010 *A Palaeographic Study of Early Writing in Egypt* (Orientalia Lovanensia Analecta 195). Leuven: Peeters.

Sbonias, K.

1995 *Frühkretische Siegel. Ansätze für eine Interpretation der sozial-politischen Entwicklung auf Kreta während der Frühbronzezeit* (BAR International Series 620). Oxford: Tempus Reparatum.

2010 Seals from the cemetery. In A. Vasilakis and K. Branigan (eds.), with contributions by T. Campbell-Green, T. Carter, D. Evely, J. Francis, F. Michelaki, K. Sbonias and S. Triantaphyllou, *Moni Odigitria. A Prepalatial Cemetery and its Environs in the Asterousia, Southern Crete*: 201–25. Philadelphia: INSTAP Academic Press.

Tsipopoulou, M. and E. Hallager
2010 *The Hieroglyphic Archive at Petras, Siteia* (Monographs of the Danish Institute at Athens 9). Athens: Danish Institute at Athens.
Ventris, M.G.F. and J. Chadwick
1973 *Documents in Mycenaean Greek*. 2nd Edition. Cambridge: Cambridge University Press.
Weilhartner, J.
2015 The design of Linear B logograms: palaeographic traditions and visual inspiration. In J. Weilhartner and F. Ruppenstein (eds.), *Tradition and innovation in the Mycenaean palatial polities. Proceedings of an International Symposium held at the Austrian Academy of Sciences, Institute for Oriental and European Archaeology, Aegean and Anatolia Department, Vienna, 1–2 March, 2013*: 255–75. Vienna: Verlag der Österreichischen Akademie der Wissenschaften.
Weingarten, J.
2003 A tale of two interlaces. In Y. Duhoux (ed.), *Briciaka. A tribute to W.C. Brice* (Cretan Studies 9): 285–99. Amsterdam: Adolf M. Hakkert.
2007 Chapter 4. The small finds. Noduli, sealings and a weight from deposits A and E. In C. Macdonald and C. Knappett, *Knossos. Protopalatial Deposits in Early Magazine A and the South-West Houses* (British School at Athens Supplementary Volume 41): 134–39. London: British School at Athens.
Willi, A.
2008 Cows, houses, hooks: the Graeco-Semitic letter names as a chapter in the history of the alphabet. *Classical Quarterly* 58: 401–23.
Woods, C.
2010 The earliest Mesopotamian writing. In C. Woods, G. Emberling and E. Teeter (eds.), *Visible Language. Inventions of Writing in the Ancient Middle East and Beyond* (Oriental Institute Museum Publications 32): 33–50. Chicago: The Oriental Institute.
Yule, P.
1980 *Early Cretan Seals: a Study of Chronology* (Marburger Studien zur Vor- und Frühgeschichte 4). Mainz am Rhein: Philipp von Zabern.